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Designing framework of hybrid photovoltaic-biowaste energy system with hydrogen storage considering economic and technical indices using whale optimization algorithm

Hongyue Sun [a], Abdol Ghaffar Ebadi [b], Mohsen Toughani [c], Saber Arabi Nowdeh [d], Amirreza Naderipour [e, *], Aldrin Abdullah [e]

[a] School of Control and Mechanical Engineering, Tianjin Chengjian University, Tianjin, 300384, China
[b] Department of Agriculture, Jouybar Branch, Islamic Azad University, Jouybar, Iran
[c] Department of Fishery, Islamic Azad University Babol Branch, Babol, Iran
[d] Golestan Technical and Vocational Training Center, Gorgan, Iran
[e] School of Housing, Building and Planning Universiti Sains Malaysia, 11800, Penang, Malaysia

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ABSTRACT

In this study economic, reliable and environmentally friendly designing of a hybrid photovoltaic-biowaste-fuel cell (PV-Biowaste-FC) system based on hydrogen storage energy is presented using whale optimization algorithm (WOA) considering the availability of components for 20 years useful lifespan of the project. The WOA is a robust meta-heuristic method for solving optimization problems with high convergence speed and accuracy. The fuel cell system includes an electrolyzer, a hydrogen storage tank, and a fuel cell stack. The objective function is defined as minimization of total net present cost (TNPC) include investment, maintenance and repair and replacement cost and reliability constraint is considered as loss of power supply probability (LPSP). The optimization variables include the area occupied by photovoltaic (PV) panels, number of biowaste units, electrolyzers, hydrogen storage tanks, fuel cells, and inverters that are optimally determined by the WOA considering the TNPC and LPSP. To validate the WOA method in the PV-Biowaste-FC designing, its performance is compared with the particle swarm optimization (PSO) method. Simulations have been carried out for different scenarios including designing of Biowaste-FC and PV-Biowaste-FC system, evaluating the effect of PV, Biowaste and inverter availability and also effect of investment cost of FC on the system designing. Simulation results show that the hybrid system with the participation of all resources and hydrogen storage system as the PV-Biowaste-FC presents the minimum TNPC (2.820 M$) and the lower LPSP (0.0029) compared to the other combinations. Also, the cost of energy (COE) value for the hybrid PV-Biowaste-FC system is obtained at 0.5238 $/kWh. The proposed method for designing the hybrid system provides better performance than the PSO in achieving less TNPC and better reliability with higher convergence speed and accuracy. Further, the effects of the availability of PV units, biowaste, and inverter on the system designing are evaluated. According to the results, decreasing the availability of the components increases the TNPC and weakens the reliability indices.

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1. Introduction

In recent years, the growth in energy demand has led to significant consumption of fossil fuels and an inevitable increase in energy prices. Concerning the rise in energy prices, the exhaustible resources of fossil fuels, and air pollution, the tendency to utilize renewable energy sources (RESs) such as photovoltaic (PV), wind, biowaste, geothermal, tidal, and hydroelectric power has grown significantly [1,2]. In a country such as Iran, fossil fuel resources, as well as RESs such as PV and biowaste, have great potential [3]. Therefore, the country's energy policy should be such that it does not rely solely on fossil fuels to provide the necessary conditions for the exploitation of RESs. Due to the vastness of Iran and the diversity of potential energy in its different regions, off-grid energy

* Corresponding author.
E-mail address: namirreza@utm.my (A. Naderipour).

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systems are quite desirable to supply remote loads from the power grid. Fluctuations in their output power due to the intermittent nature of RESs, as well as high investment costs, are important challenges when employing this type of energy resources. Therefore, in order to achieve a reliable and cost-effective system, different energy sources are combined to cover each other's power fluctuations [2]. Hybrid systems also use storage devices to manage energy between energy generation resources and the consumer. The purpose of designing hybrid systems is to determine the optimal size of the system equipment taking into account economic and technical indices. The economic index reflects the energy generation costs of the system and the technical index addresses the load supply reliability of the system. Thus, the design aims to determine the optimal size of the equipment in such a way that the costs of the system are minimized and also the load of the system is supplied with the desired reliability. On the other hand, achieving optimal equipment size requires the use of robust intelligent optimization methods that can achieve maximum reliability in addition to minimizing costs [4–6].

Numerous studies have been conducted on the design of hybrid sustainable energy systems for remote loads from the grid. In [7], the determination of the optimal size of the battery bank and PV panels of a wind-PV system has been investigated. The number of PV panels and batteries is presented based on cost analysis and specific load supply. The results showed that the optimal combination of PV arrays and batteries depends on the load profile and reliability of the hybrid system corresponding to the lowest cost and more reliability. In [8], probabilistic design and placement of wind resources are presented via a modified spotted hyena optimizer. In [9], the optimal design of a hybrid wind-PV system to supply a remote radio station is presented. The results highlight that the use of PV arrays alone is not technical and economical because in this case a large amount of PV arrays and batteries are needed and this would not be economically viable. So combination of PV and wind resources integrated battery storage is optimal option to supply the load. In [10], a multi-criteria optimization is presented to solve the wind resources allocation in the network with aim of losses reduction and voltage improvement. The results showed that the optimal size of the wind resources causes the losses and voltage deviation minimization. In [11], an energy model is presented based on the optimal design of a PV system with hydrogen storage. The purpose of optimization is to find the optimal size of the equipment to minimize the energy costs of the system taking into account the LPSP index. System costs include the costs of investment, maintenance and repair, and replacement. The results showed that the proposed energy model based on PV renewable sources based on hydrogen storage has reduced energy generation costs and load supply by achieving the desired reliability. In [12], the optimal design of a hybrid PV-wind-battery system is developed based on the balance between supply and demand to minimize costs. The shortage of load power is considered as an index to examine the capability of the hybrid system in supplying the demand. Further, the optimal design of a wind-PV system is presented in the form of a single-objective optimization problem. The battery storage system has been used to compensate for the system's power shortage and increase reliability. This study intends to minimize system power generation costs, including the cost of the equipment used, including PV panels, wind turbines, and batteries. The results showed the suggested framework of the hybrid system designing can be determined the optimal sizing of the system with achieving the lowest cost with high reliability. A model has been presented in [13] for the optimal design of a hybrid PV-wind system with fuel cell storage via sine–cosine algorithm. This research conducted minimizes energy generation costs and supplies the load properly. The optimal size of the equipment is calculated taking into account the cost and reliability indices. The results showed that when the capacity of the hydrogen increases, the capacity of the PV arrays decreases, and the system costs increase but the system reliability is weakened. Authors in [14] present the optimal design of a hybrid wind-PV-diesel generator with a battery storage system to minimize the annual costs of the system, the LPSP, and fuel pollution costs using a multi-objective quasi-evolutionary algorithm. The results cleared that with the contribution of the battery and diesel as storage and backup units respectively, optimal sizing of the system is obtained with lower cost and more reliability. In [15], the design of a wind-PV system with a battery storage system is presented to minimize the cost using a grey wolf algorithm taking into account the energy not supplied index. The results show that system costs increase when it needs to supply a larger amount of load and also the best combination with lower cost is the wind-PV-battery system. In [16], the optimal design of a wind-PV-battery system is performed as grid-connected based on the reliability constraint and with aim of minimizing the net present cost and CO_2 cost considering not-supplied energy probability. The results showed that the reliability is improved and the system cost is reduced in grid-connected condition. The optimal design of a hybrid wind-PV-diesel-battery system is presented in [17] to minimize TNPC taking into account the overall energy shortage of the system. The optimal combination of system equipment is obtained based on the minimum cost. The results showed using of the battery and diesel as storage and backup units respectively, reduces the energy generation cost and improves reliability. In [18], a hybrid wind-PV system with a fuel cell storage system has been used to supply an off-grid load. This study presents the optimal design of the system by minimizing life span costs and considering the load reliability index as load interruption probability using an improved sine–cosine algorithm (ISCA). In [19], the technical and economic design of a hybrid wind-PV-fuel cell system is studied considering load interruption probability and minimizing lifespan cost of the hybrid system using hybrid grey wolf optimizer-sine cosine algorithm with the determination of the best system combination for remote application. The results cleared the storage system based hydrogen can be compensated the deficit power of the generation resources and improve the load reliability. Moreover, in recent years, there have been many ways to provide methods for predicting data. Moreover, root mean square propagation (RmsProp) and adaptive moment estimation (Adam) belong to gradient descent optimization algorithms, which have also been used for optimization. The RMSProp is an adaptive stochastic algorithm to train deep neural network. The RMSProp discards previous and preserves only current knowledge on the gradient. Also, adaptive moment estimation (Adam) is a method of SGD optimization that the learning rate is measured for each parameter. The upgrade operation considers the smooth gradient variant and provides a bias correction mechanism. So far there is no method in gradient descent optimization algorithms that reflects population-based algorithms such as evolutionary and swarm algorithms proven powerful in many optimization tasks and swarm optimization algorithms do not require gradient information [20].

By examining prior studies in view of the design of hybrid systems, it becomes obvious that in most studies, it is recommended to use methods with higher optimization and computational power to design the system more accurately and achieve lower cost and higher reliability. As in previous studies, the focus of design studies has been on achieving the best system configuration at the lowest cost. The performance probability of system equipment is an important issue in the precise design of hybrid systems that have not been well addressed in previous studies. The performance probability of equipment depends on its availability. The higher

availability of the equipment causes a higher performance probability. Also, the availability of equipment is related to their failure and repair rates. Therefore, one of the challenges of previous studies is not evaluating this concept. The literature review showed that the availability of RES equipment and inverters in the design of such systems has not been carried out so far. Moreover, the effect of biowaste energy exploitation on hybrid PV-wind systems has not been comprehensively investigated yet. Because disposing of municipal waste, biowaste can also generate electricity, is always and widely available as a renewable energy source, is less expensive than fossil fuels and biowaste adds revenue for manufacturers. In addition to these concepts, presenting a new design framework based on economic evaluation (cost) and technical analysis (reliability) by providing desirable indices of reliability considering availabilities of photovoltaic and biowaste resources is one of the important goals.

Moreover using fuel cell based on hydrogen is suggested as a storage system to compensate the deficit power of the hybrid system to supply the load, continuously. Batteries serve as short-term storage and fuel cells as long-term storage. Batteries and capacitors lose the energy stored in them over time and need to be periodically recharged even without use. Also, the fuel cell compared to batteries can save money. They eliminate the need to change, charge, and manage batteries, subsequently reducing labor, time, and peak power demand. Moreover, the reliability of fuel cell is more than the batteries [21].

The designing framework of a hybrid energy system that consists of PV panels, biowaste, and fuel cell (PV-Biowaste-FC) is presented based on hydrogen energy to supply the electricity demand of an islanded area considering technical and economic indices for Ardabil city in Iran country (latitude: 38° 17′, longitude: 48° 15′, altitude: 1345 m). One of the state-of-the-art technologies for energy extraction of biowaste resources is based on biological processes, which is one of the final products of this biogas process and can be burned to generate electricity and heat [22]. In the hybrid PV-Biowaste-FC system, PV and biowaste are operated as the main energy generation sources and also the FC based on the hydrogen energy is considered as the storage system to improve the load reliability. In this research, the whale optimization algorithm (WOA) [23,24] is applied to design the hybrid system, optimally. The WOA is inspired by the hunting behavior of humpback whales based on the formation of circular bubbles and has been a robust approach to solve optimization problems with high optimization speed and accuracy. The bubble-net behavior of a whale is similar to a spiral movement between the whale and prey positions used to model this algorithm. Moreover, the other advantages of the WOA are simplicity in implementation and prevent premature convergence [23,24]. The objective function of optimization is to minimize the TNPC as the economic index of the design. TNPC is considered as the initial investment cost, maintenance costs, and replacement costs for PV, biowaste, and fuel cell units. Also, a technical index as a reliability constraint is defined as the LPSP. The optimization variables include the total area occupied by PV panels, the number of biowaste units and electrolyzers, the number of hydrogen tanks, and the number of fuel cells, which are given according to the objective function, and the reliability constraint is optimally determined. The study also presents the optimal design of the hybrid system and determines the optimization variables in different system combinations. Moreover, the effect of the equipment availability on optimization variables and technical and economic indices has been investigated for the best combination of the energy system. Additionally, the effects of inverter availability and increase of the initial investment cost of fuel cells on the design have been investigated.

2. Understudy hybrid system

2.1. Operation of the hybrid system

A hybrid system including PV panels, biowaste, and fuel cells has been used to supply a remote load. Also, to continuously supply the load and increase the reliability of the supply, the current research employs a fuel cell system including an electrolyzer and a hydrogen storage tank as an energy storage system. The system under study is shown in Fig. 1.

The energy management strategy (EMS) is vital to operate the hybrid system to meet the system demand. The EMS is a strategy that is possible to conduct the energy units in time duration to meet the system demand. Based on EMS the energy units as PV and wind units clear suitable expletive to supply the system demand with a designing framework based on TNPC minimization and observance of the LPSP. The EMS is presented as follows [18,19]:

- If the total generated power by the PV and biowaste is equal to the load demand, all the power is injected into the load through the inverter.
- If the total generated power by PV and biowaste is greater than the load demand, the excess power is delivered to the electrolyzer for hydrogen production. If the injected power to the electrolyzer exceeds its rated capacity or the hydrogen storage tank is full, the excess power is lost in a dump load.
- If the total generated power by the PV and biowaste is less than the demand, the load is supplied by the FC. If the FC is not capable to supply the load fully, part of the load is interrupted, which causes the loss of power supply probability (LPSP) as the reliability index.

2.2. Modeling of the hybrid system

2.2.1. Modeling of PV panels

The output power of the PV panel is defined using solar irradiance, ambient temperature, and efficiency of the PV module [17–19]:

$$P_{pv}(t) = A_{pv}.\eta_{pv}(t).I(t) \tag{1}$$

where P_{PV} is the power generated by the PV array (kW), A_{PV} is the area occupied by PV panels (m^2), η_{pv} is the instantaneous efficiency of PV panels, and I denotes the solar irradiance (W/m^2) available on the ramp.

Instantaneous efficiency of the PV panel is found as follows [17–19]:

$$\eta_{pv}(t) = \eta_r.\eta_t \times \left[1 - \beta(T_a(t) - T_r) - \beta.I(t).\left(\frac{NOCT - 20}{800}\right).(1 - \eta_r.\eta)\right] \tag{2}$$

where η_r is the reference efficiency of PV panels, η_t is the power tracking efficiency, β is the gain heat coefficient (between 0.004 and 0.006) [18]. Also, T_a shows the ambient temperature (°C), T_r is the reference heat temperature of the PV cell (°C), and NOCT is the Nominal Operating Cell Temperature (°C).

2.2.2. Modeling of biowaste power generation

Biowaste is obtained from sources such as plants and animals that can be replaced. Plants can be grown over and over again on a plot of land at no extra cost. Therefore, raw materials are easily available and useable, and as a result, are good sources of renewable

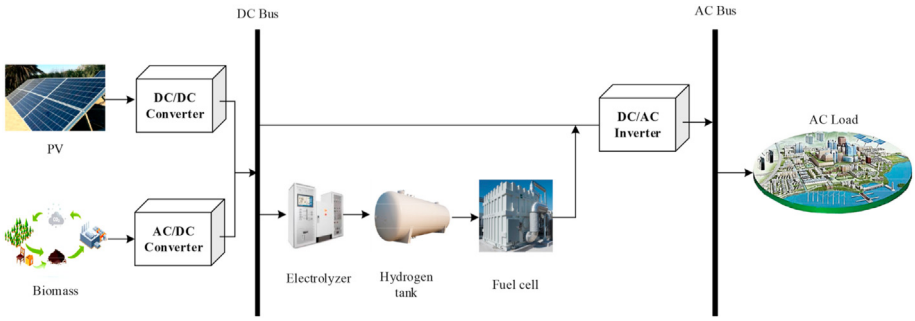


Fig. 1. Schematic of the proposed hybrid PV-Biowaste-FC system.

energy. Biowaste uses waste and converts it into an energy source. Therefore, it reduces the waste disposal process and the resulting pollution caused by them.

The amount of biowaste energy is calculated based on the generation flow speed, temperature, electrical conversion efficiency and also lower heating value of the gas. The output generated power by biowaste ($P_{e,\,bg}$) is defined as follows [22]:

$$P_{e,bg}(t) = \eta_{e,bg}.Q_{bg}(t).LHV_{bg} \tag{3}$$

where $\eta_{e,\,bg}$ is the electrical conversion efficiency (between 25 and 40 %), $Q_{bg}(t)$ is the gas flow (m^3/h) at time t, and LHV_{bg} shows the lower heating value of the gas (M^3/kWh) [22].

Biogas heating value according to the percentage of methane in the gas and lower heating value of methane is obtained as follows [22]:

$$LHV_{bg} = \frac{P_{CH_4}}{100}.LHV_{CH_4} \tag{4}$$

where P_{CH4} represents the percentage of methane in the gas and LHV_{CH4} is the lower heating value of methane (36.3 MJ/m or 3–10 kW h/m^2) [22].

Gas flow based on the operating power of the biowaste generator is defined based on the power based on which the biowaste generator operates [22].

$$Q_{bg}(t) = \frac{PW_{bg}(t)}{\eta_{e,bg}.LHV_{bg}} \tag{5}$$

where $PW_{bg}(t)$ indicates the operating power of the biowaste generator at time *t*.

2.2.3. Modeling of the fuel cell system

The fuel cell system includes an electrolyzer, a hydrogen storage tank, and a fuel cell stack. Hydrogen is extracted from water using the electrolysis process by the electrolyzer Sand is stored in a high-pressure hydrogen tank. In the case of a shortage of hydrogen power, it is injected into the fuel cell to generate power.

- **Electrolyzer**

The output power of the electrolyzer is given by [4]:

$$P_{el-\ tank} = P_{ren-el} \times \eta_{el} \tag{6}$$

where η_{el} is the efficiency of the electrolyzer and $P_{ren\text{-}el}$ is the power injected into the electrolyzer by PV panels and biowaste.

- **Hydrogen storage tank**

Energy storage in the hydrogen storage tank at time *t* is given as [18,19,25]:

$$E_{tank}(t) = E_{tank}(t-1) + P_{el-tank}(t) \times \Delta t - P_{tank-FC}(t) \times \Delta t \times \eta_{storage} \tag{7}$$

where Δt is the time step (1 h) and $\eta_{storage}$ is the efficiency of the storage system (95 %). Mass of the hydrogen within the tank is defined as:

$$m_{storage}(t) = \frac{E_{storage}(t)}{HHV_{H_2}} \tag{8}$$

where HHV_{H2} represents the higher heating value of hydrogen [4].

The energy of hydrogen in the tank is always bounded as follows:

$$E_{\tan\ k,\min} \le E_{\tan\ k}(t) \le E_{\tan\ k,\max} \tag{9}$$

- **Fuel cell stack**

The output power of the fuel cell stack is calculated by [4,18]:

$$P_{FC-\ inv} = P_{tank-FC} \times \eta_{FC} \tag{10}$$

where η_{FC} shows the efficiency of the fuel cell.

2.2.4. Modeling of the inverter

Power injected by the inverter to the load is obtained based on the efficiency of the inverter:

$$P_{inv-\ load} = (P_{FC-inv} + P_{ren-inv}) \times \eta_{inv} \tag{11}$$

3. Modeling of system reliability

The most important intention behind using the PV-Biowaste-FC system is to increase the reliability of load supply, in addition to reducing energy generation costs [26]. Conducting the calculations of the reliability has to be considered along with economic evaluations (cost) of energy generation by the hybrid system. Three reliability indices including loss of power supply probability (LPSP), loss of load expectation (LOLE), and loss of energy expectation (LOEE) are used to evaluate the reliability of the hybrid system. These are formulated in the following subsections.

3.1. Reliability indices

The LPSP index is stated when energy generation sources with storage cannot fully supply the load and part of the load is interrupted. In other words, the LPSP represents the ratio of the load not-supplied to the total load of the system [27]:

$$LPSP = \frac{ENS}{\sum_{t=1}^{T} E_L(t)} \tag{12}$$

Where, $\sum_{t=1}^{T} E_L(t)$ is the total demand and *ENS* represents the energy not-supplied by the system. Also, T refers to the study period during a year as 8760 h. Indices such as LOLE and LOEE are calculated as follows:

$$LOLE = \sum_{t=1}^{T} E[LOL(t)] \tag{13}$$

where $E[LOL(t)]$, the mathematical expectation of load interruption at time *t*, is calculated using (14):

$$E[LOL] = \sum_{s \in S} T_s \times P_s \tag{14}$$

here, P_s is the probability of occurrence under conditions *s*, T_s shows the load interruption time, and S represents the whole situations established in the hybrid system.

$$LOEE = EENS = \sum_{t=1}^{T} E[LOE(t)] \tag{15}$$

where $E[LOE(t)]$ is the mathematical expectation of the loss of energy at time *t* given by:

$$E[LOE] = \sum_{s \in S} Q_s \times P_s \tag{16}$$

where Q_s is the amount interrupted load (kWh) if conditions *S* occur.

3.2. Reliability model of energy generation

This study evaluates the reliability of the hybrid system considering the forced outage rate (FOR) of PV and biowaste units. The probability of outage of n_{Bio}^{fail} number of biowaste units from all installed biowaste units (N_{Bio}) and the outage of n_{PV}^{fail} number of PV units from all installed PV panels (N_{PV}) is defined using (17) [4]:

$$f_{ren}\left(n_{Bio}^{fail}, n_{PV}^{fail}\right) = \left[\binom{N_{Bio}}{n_{Bio}^{fail}} \times AV_{Bio}^{N_{Bio}-n_{Bio}^{fail}} \times (1-AV_{Bio})^{n_{Bio}^{fail}}\right] \times \left[\binom{N_{PV}}{n_{PV}^{fail}} \times AV_{PV}^{N_{PV}-n_{PV}^{fail}} \times (1-AV_{PV})^{n_{PV}^{fail}}\right] \tag{17}$$

Where, AV_{Bio} and AV_{PV} show the availability of biowaste and PV units, respectively.

If n_{PV}^{fail} number of PV panels and n_{Bio}^{fail} number of biowaste units experience an outage, the output power generation based on PV and biowaste units is given by:

$$P_{ren}\left(n_{PV}^{fail}, n_{Bio}^{fail}\right) = \left(N_{PV} - n_{PV}^{fail}\right) \times P_{PV} + \left(N_{Bio} - n_{Bio}^{fail}\right) \times P_{Bio} \tag{18}$$

According to Eq. (18) and the probability of outage of PV and biowaste units, the mathematical expectation of generated energy will be:

$$E[P_{ren}] = \sum_{n_{Bio}^{fail}}^{N_{Bio}} \sum_{n_{PV}^{fail}}^{N_{PV}} \left\{P_{ren}\left(n_{Bio}^{fail}, n_{PV}^{fail}\right) \times f_{ren}\left(n_{Bio}^{fail}, n_{PV}^{fail}\right)\right\} \tag{19}$$

Therefore, the output energy generation model of the hybrid system based on PV and biowaste units considering their availability is given as follows:

$$E[P_{ren}] = N_{PV} \times P_{PV} \times AV_{PV} + N_{Bio} \times P_{Bio} \times AV_{Bio} \tag{20}$$

4. Objective function and problem constraints

4.1. Objective function

This study aims to determine the optimal size of the equipment in a hybrid PV-Biowaste-FC system, which includes the number of biowaste, the area occupied by PV panels, number of electrolyzers, number of hydrogen tanks, number of fuel cells and inverters while minimizing energy generation costs based on the WOA method. The following provides the formulation of the objective function.

4.1.1. Formulation of the objective function

The objective function of the optimization problem is defined as the minimization of the TNPC of the equipment in the hybrid PV-Biowaste-FC system:

$$ObjectiveFunction = Min.TNPC\left(A_{pv}, N_{bg}, N_{PV}, N_{el}, N_{\tan k}, N_{FC}\right) = \sum_{i=pv,pg,el,\tan k,fc} TNPC_i = N_i \times \left(\begin{array}{l} C_{capital,i} + C_{replacement,i} \times \sum_{n=1}^{y_i} \frac{1}{(1+ir)^{n.L_i}} \\ \\ + C_{maintenance,i} \times \frac{(1+ir)^R - 1}{ir(1+ir)^R} \end{array} \right) \quad (21)$$

where i represents the equipment, N is the number of units of the size of the equipment, $C_{capital,i}$ is the initial investment cost ($/Unit), $C_{replacement,i}$ gives the replacement cost ($/Unit), $C_{maintenance,i}$ is the annual maintenance and repair cost ($/Unit), and R is the duration of the project (20 years). Also, ir is the real interest (6 %) based on the nominal interest ir_{nominl} and the annual inflation rate (f). In the above equation, y and L represent the number of replacements and the useful lifelong of the equipment, respectively. Thus, the aim of designing the hybrid system is to minimize $TNPC$ considering the reliability of LPSP.

4.2. Constraints

The objective function should be optimized considering the following constraints [4,17–19]:

$$0 \le LPSP \le LPSP_{\max} \quad (22)$$

$$0 \le N_i \le N_i^{\max} \quad (23)$$

$$E_{\tan k}(0) \le E_{\tan k}(8760) \quad (24)$$

In the above constraints, $LPSP_{\max}$ is the maximum value of the reliability of LPSP and $N_i^{\max}$ gives the maximum number of equipment in the system. The maximum and minimum number of biowaste units, the area occupied by PV panels, the number of electrolyzers, hydrogen tank, and inverter are given in Table 1.

5. Proposed optimization algorithm

5.1. Introducing the WOA

Whales are the largest mammals and are knows as smart and experienced hunters [23,24]. Fig. 2 shows the hunting behavior of whales based on the bubble-net foraging method that includes two maneuvers: upward spirals and double loops.

The WOA algorithm is formulated based on prey encircling using circular bubbles, foraging maneuver, and searching. The following describes the mentioned steps.

5.1.1. Encircling prey

The whale encircles the prey after discovering the prey's location. Since the prey is located within the search space and its precise location is not exactly specified, the WOA selects the best solution in the nearest location to the prey. Then, position updating is carried out with respect to the best search agent [23].

Table 1
Maximum and minimum values of designing constraints.

Value/Item	$N_{Biowaste}$	$A_{PV}(m^2)$	N_{el}	$N_{\tan k}$	N_{FC}	N_{inv}	$LPSP_{\max}$
Maximum value	300	700	500	500	100	100	1
Minimum value	0	0	0	0	0	0	0

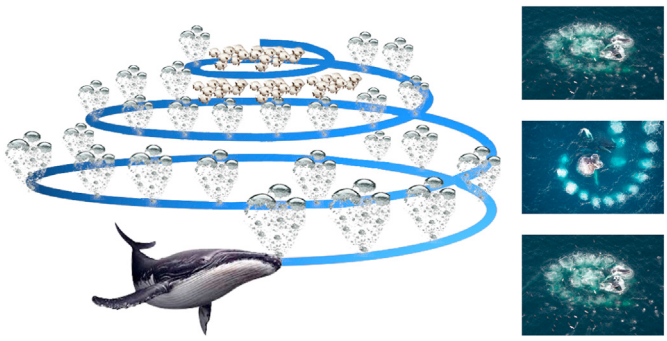


Fig. 2. Bubble-net foraging behavior of humpback whales [23].

$$\vec{D} = \left| \vec{C}.\vec{X^*}(t) - \vec{X}(t) \right| \quad (25)$$

$$\vec{X}(t+1) = \vec{X^*}(t) - \vec{A}.\vec{D} \quad (26)$$

where t is the current iteration, A and C are coefficient vectors, X* is the position of the best solution, X represents the position vector of the prey, and || is the absolute value. Vectors A and C are defined as follows [23]:

$$\vec{A} = 2\vec{a}.\vec{r} - \vec{a} \quad (27)$$

$$\vec{C} = 2.\vec{r} \quad (28)$$

where $\vec{a}$ is a decreasing linear vector from 2 to 0, and $\vec{r}$ represents a random vector in the range of [0, 1].

5.1.2. Bubble-net attacking method (exploitation phase)

The bubble-net behavior of a whale is similar to a spiral movement between the whale and prey positions. By swimming around the prey, the whale travels a shrinking circular path and simultaneously a spiral movement is given as follows [23]:

$$\vec{X}(t+1) = \left\{ \begin{array}{l} \vec{X^*}(t) - \vec{A}.\vec{D} \text{ if } p < 0.5 \\ \vec{D'}.\, e^{bl}.\cos(2\pi l) + \vec{X^*}(t) \text{if } p \ge 0.5 \end{array} \right\} \quad (29)$$

Where, l is a random number in [-1,1] and p refers to a random number in [0,1].

5.1.3. Searching the prey (exploration phase)

The position of the search agent in this phase is updated based on the search agent and is considered as the best search agent, the model of which is provided here [23]:

$$\vec{D} = \left| \vec{C}.\vec{X_{rand}} - \vec{X} \right| \quad (30)$$

$$\vec{X}(t+1) = \vec{X_{rand}} - \vec{A}.\vec{D} \quad (31)$$

where $\vec{x_{rand}}$ denotes the random position vector.

At each iteration of the algorithm, the search agent randomly considers its position as the best solution and updates it. The structure of exploration and exploitation decreases the value of $\vec{a}$ parameter from 2 to 0. According to the value of p in Eq. (29), the WOA can switch between each spiral paths or circular movement. Finally, the algorithm is terminated after satisfying the convergence criterion and achieving the best fitness.

5.2. Implementation of the WOA

The implementation trend of the WOA in solving technical and economic design problem of the hybrid PV-Biowaste-FC system is described as follows.

Step 1. Design data are input to the program. The data of irradiance, temperature, and gas flow of the biowaste, technical and economic data of the system equipment, as well as the maximum number of iterations and the population of the algorithm, are input to the optimization program.

Step 2. The initial population of decision variables is generated randomly with respect to decision variables.

Step 3. The objective function of the problem is calculated for the set of variables selected in Step 2, is calculated to evaluate the problem constraints. The best whale with the best variable set and the best objective function is determined.

Step 4. The whale populations are updated and decision variables are selected randomly for the new population. After the objective function is calculated, the corresponding whale is determined with the best set of variables and the minimum value of the objective function. If it is better than the objective function obtained in Step 3, it will be replaced.

Step 5. The previous steps continue until the algorithm converges to the best value of the objective function and performs the maximum number of iterations of the algorithm. Then, after determining the optimal variable set with the minimum cost, the algorithm is terminated.

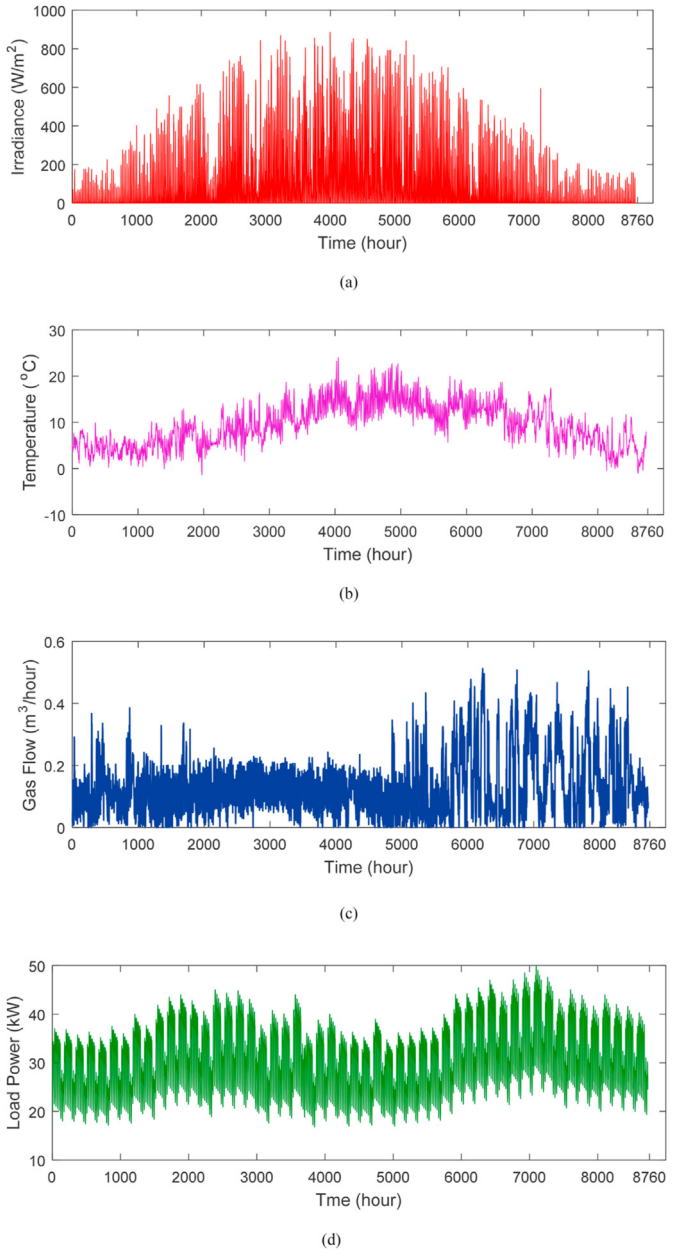


Fig. 3. Annual profile of a) irradiance b) ambient temperature c) gas flow and d) IEEE load demand [4,28].

6. Simulation results and discussion

Curves related to irradiance, ambient temperature, biowaste gas flow, as well as IEEE load demand profile over a year are given in Fig. 3. The data of irradiance and ambient temperature are related to Nowdeh Malek region in Iran country located in latitude: 36°88′ and longitude: 54°65′[28]. The annual load curve load profile of the IEEE standard is considered with a 50 kW peak value equal to 269.15 MWh [4]. The parameters of the equipment in the hybrid PV-Biowaste-FC system are also listed in Table 2 and the assumptions are given in Table 3. In this research, MATLAB area (R2015a) is applied to implement the methodology based on a personal computer with Core i7, 3.1 GHz CPU, 8 GB memory, 1 T HDD and Windows 7 operating system. Based on the trial and error test, the maximum number of iterations and the number of populations are set at 200 and 50, respectively. The average computational cost for WOA and PSO for each repetition (30) is obtained at 219 and 228 min, respectively for achieving the designing results of the hybrid PV-Biowaste-FC system.

The following scenarios are presented to evaluate the hybrid PV-Biowaste-FC system.

Scenario 1. Design of the hybrid Biowaset-FC system using WOA method and comparing its performance with the PSO method.

Scenario 2. Design of the hybrid PV-Biowaste-FC system using WOA and PSO methods.

Scenario 3. Design of the hybrid PV-Biowaste-FC system for different availabilities of PV and biowaste units using the WOA method.

Scenario 4. Design of the hybrid PV-Biowaste-FC system for different availabilities of the inverter using the WOA method.

Scenario 5. Evaluating the effect of increasing the initial investment cost of fuel cell on the design of the hybrid PV-Biowaste-FC system using the WOA method.

6.1. Simulation results: scenario 1

This section designs a hybrid Biowaste-FC system using the WOA method. Also, in order to evaluate the performance of the proposed method, the problem has been compared and analyzed with the PSO. In this section, decision variables include the number of biowaste units, number of electrolyzers, number of hydrogen tanks, number of fuel cells, and number of inverters. The convergence curves of the WOA and PSO algorithms in solving the optimization problem are shown in Fig. 4. As shown in this figure, the WOA method, similar to the robust PSO method, achieves almost the same value as the objective function. Therefore, the performance of the WOA method has also been optimal in terms of cost.

Table 4, tabulates the optimal results of the decision variables obtained by WOA and PSO methods. According to the optimal variables obtained by the WOA method, TNPC values and reliability

Table 2
Parameters of the equipment in the hybrid PV-Biowaste-FC system [4,17–19].

Equipment	Initial investment cost (US$/unit)	Replacement cost (US$/unit)	Annual maintenance and repair cost (US$/unit-yr)	Useful lifelong (year)	Efficiency (%)	Capacity	Availability (%)
Biowaste	6500	6500	325	20	–	1 kW	100
PV panel	519	519	20	20	–	1 m^2 (1 kW)	100
Electrolyzer	2000	1500	25	20	75	1 kW	100
Hydrogen tank	1300	1200	15	20	95	1 kg	100
Fuel cell	3000	2500	175	5	50	1 kW	100
Inverter	800	750	8	15	90	1 kW	100

Table 3
Assumptions for the hybrid system [4,17–19].

$LPSP_{max}$	Real interest rate	Useful lifelong of the system
0.01	6 %	20 years

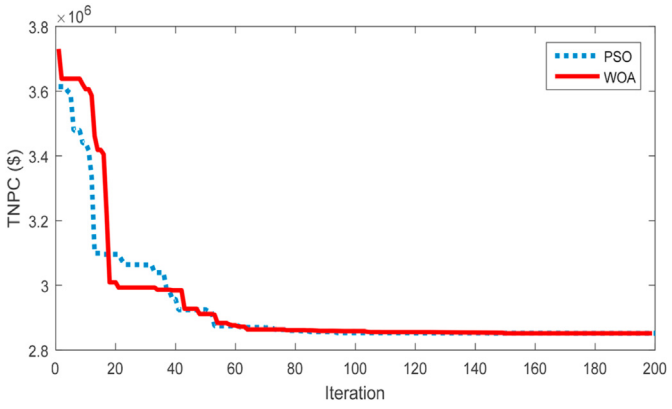


Fig. 4. Convergence curve of WOA and PSO methods in Scenario 1.

Table 4
The optimal size of the understudy system equipment in Scenario 1.

Method/Parameter	$N_{Biowaste}$	A_{PV}	N_{el}	$N_{\tan k}$	N_{FC}	N_{inv}
WOA method	189	–	62	216	31	47 (47.44)
method PSO	189	–	60	222	32	47 (47.32)

indices are given in Table 5. The cost of the system by WOA and PSO is obtained 2.850 and 2.852 M$, respectively. The value of COE or system costs incurred per kilowatt is $ 0.5294 in WOA and $ 0.5298 in PSO. It is observed that the cost obtained by WOA for each kW is less than that of the PSO method. On the other hand, it was observed that the WOA method in the hybrid Biowaste-FC system supplied a slightly better demand than the PSO method, which is due to LCOE. Also according to Table 4, the WOA is obtained better reliability indices (lowest values) than the PSO results. So the WOA is superior in hybrid Biowaste-FC system designing to the well-known PSO method. The annual energy produced by biowaste units and also hydrogen storage energy is depicted in Fig. 5.

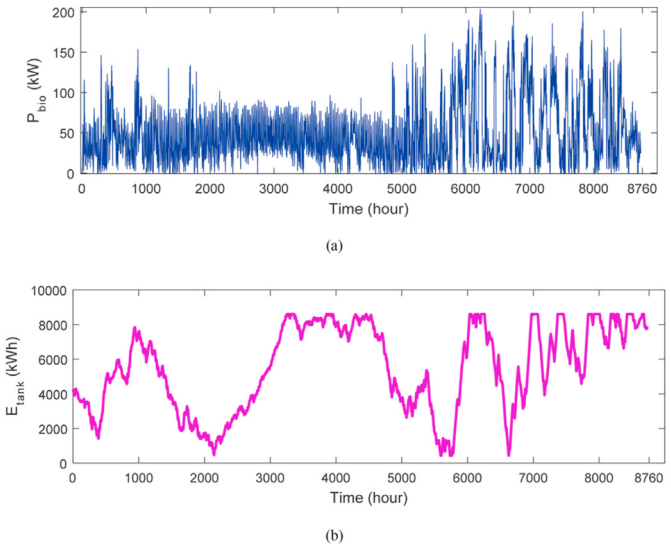


Fig. 5. Contribution of biowaste and storage in Scenario 1, a) Annual biowaste generation and b) annual hydrogen storage energy.

Moreover, the reliability indices such as LOEE, LOLE and LPSP are presented in Fig. 6. According to these figures, hydrogen storage is integrated with biowaste resources to support the demand and improve load reliability.

6.2. Simulation results: scenario 2

In this section, the results obtained from optimizing the hybrid PV-Biowaste-FC system using WOA and PSO algorithms are presented. Decision variables in the given optimization method include the area occupied by PV panels and numbers of biowaste units, electrolyzers, hydrogen tanks, fuel cells, and inverters. The convergence curve of the optimization algorithms is shown in Fig. 7. As shown in Fig. 7, the WOA method has a lower convergence tolerance than the PSO and converges in a fewer number of iterations. The value of the objective function obtained by the WOA and PSO methods is almost the same: $ 2.821 and $ 2.822, respectively. Hence, since the PSO method is a powerful tool to solve optimization problems, the WOA method has also performed well.

Table 6, provides the results of variable sizes obtained from the

Table 5
Cost values and reliability indices in Scenario 1.

Method/Parameter	TNPC (M$)	COE ($/kWh)	LPSP	LOLE (hr/yr)	LOEE (MWh/yr)
WOA method	2.850	0.5294	0.0060	295	1.615
method PSO	2.852	0.5298	0.0061	301	1.641

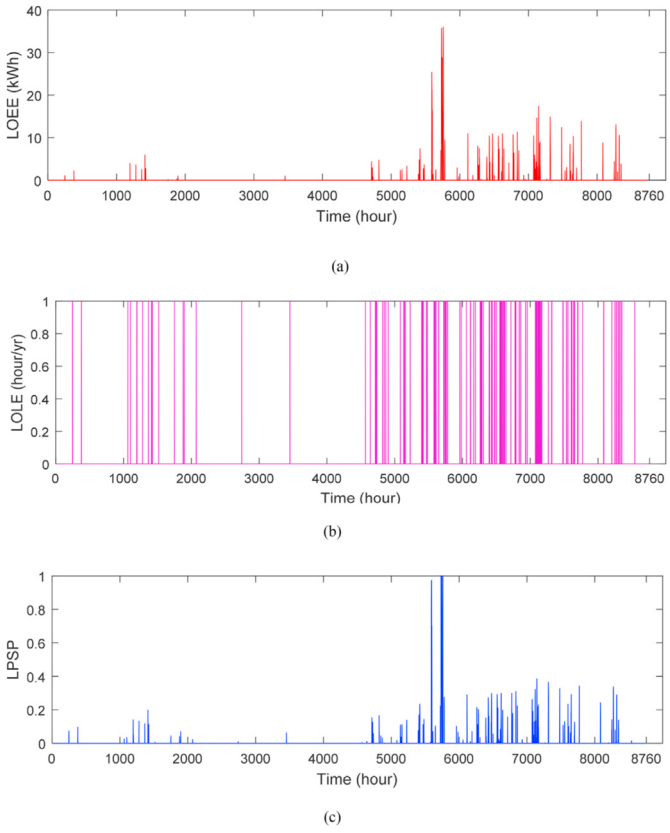


Fig. 6. The reliability indices in Scenario 1 a) LOEE b) LOLE c) LPSP.

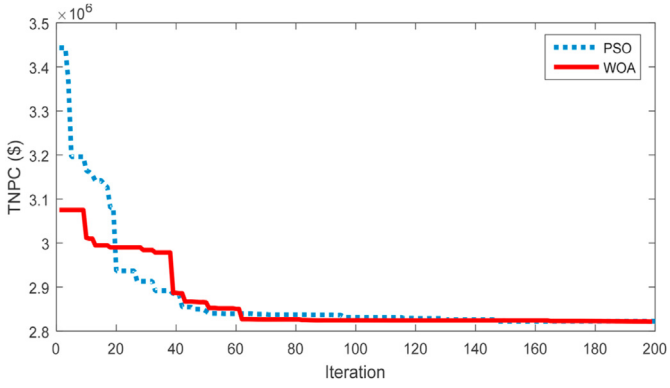


Fig. 7. Convergence curves obtained by WOA and PSO methods in Scenario 2.

Table 6
Optimal sizing of the equipment of the understudy system in Scenario 2.

Method/Parameter	$N_{Biowaste}$	A_{PV}	N_{el}	$N_{\tan k}$	N_{FC}	N_{inv}
WOA method	189	78.62	61	209	31	47 (47.44)
method PSO	187	64.7	65	203	31	47 (47.10)

optimization methods. Also, TNPC values and reliability indices are given by WOA and PSO methods are shown in Table 7. The results show that in the WOA method, the system load is supplied with better reliability indices. The TNPC is obtained 2.820 and 2.822 M$ using WOA and PSO, respectively. Also, COE value obtained using WOA and PSO is 0.5238 and 0.5242 $, respectively. So the WOA is obtained less cost of hybrid system designing in scenario 2. Moreover, the results obtained better reliability indices for WOA than the PSO with lower LPSP, LOLE and LOEE. The performance of the WOA method in supplying the system load has been better than the PSO method. Power generation by biowaste and PV units and hydrogen storage energy is depicted for a year and also with an hourly average of a day for the study period in Fig. 8. Moreover, the contribution of generation units with load variations and also tank hydrogen energy is illustrated for the first week of the year in Fig. 9. Then the annual reliability index as LSPP for Scenario 2 is depicted in Fig. 10. Figs. 8–10, showed the important role of the hydrogen storage tank in compensation of the load shortage and also improving the reliability indices.

6.3. Comparing the results of scenarios 1 and 2

In this section, the results of Scenarios 1 and 2 in exploiting energy resources are evaluated. Table 8, compares the cost values and reliability indices for different scenarios. As can be seen, the TNPC value of the PV-Biowaste-FC system is lower than the Bio-FC system. The COE value in scenario 1 and 2 is obtained at 0.5294 and 0.5238 $, respectively. Also, it is clear that the reliability indices are obtained better than the PSO results. The LPSP value is achieved 0.0060 and 0.0029 in scenarios 1 and 2, respectively. In other words, power generation units are able to meet system load demand at a lower cost by overlapping each other's power fluctuations and better reliability. As can be seen in Scenario 1, TNPC has the highest value when only PV panels with fuel cells are used to supply the load. Of course, in Scenario 2, better reliability indices are presented so that it has better LPSP, a fewer number of outage hours, and less energy lost during one year than Scenario 1. On the other hand, due to the better supply of the demand, its COE value is slightly higher than Scenario 1.

6.4. Simulation results of scenario 3

In this scenario, the effects of changes in the availability of energy generating units, i.e., PV and biowaste panels in the design of the PV-Biowaste-FC system are evaluated. In the ideal design, the availability of equipment is considered 100 %, but in the real design, 100 % availability is not possible, so the effect of availability changes in the design should also be evaluated, which has been studied. The performance probability of system equipment is an important issue in the precise design of hybrid systems that have not been well addressed in previous studies. The performance probability of equipment depends on its availability. The higher availability of the equipment causes a higher performance probability. Also, the availability of equipment is related to their failure and repair rates. The concept of availability means the probability of operation of the

Table 7
Cost values and reliability indices in Scenario 2.

Method/Parameter	TNPC (M$)	COE ($/kWh)	LPSP	LOLE (hr/yr)	LOEE (MWh/yr)
WOA method	2.820	0.5238	0.0029	220	0.974
method PSO	2.822	0.5242	0.0040	261	1.300

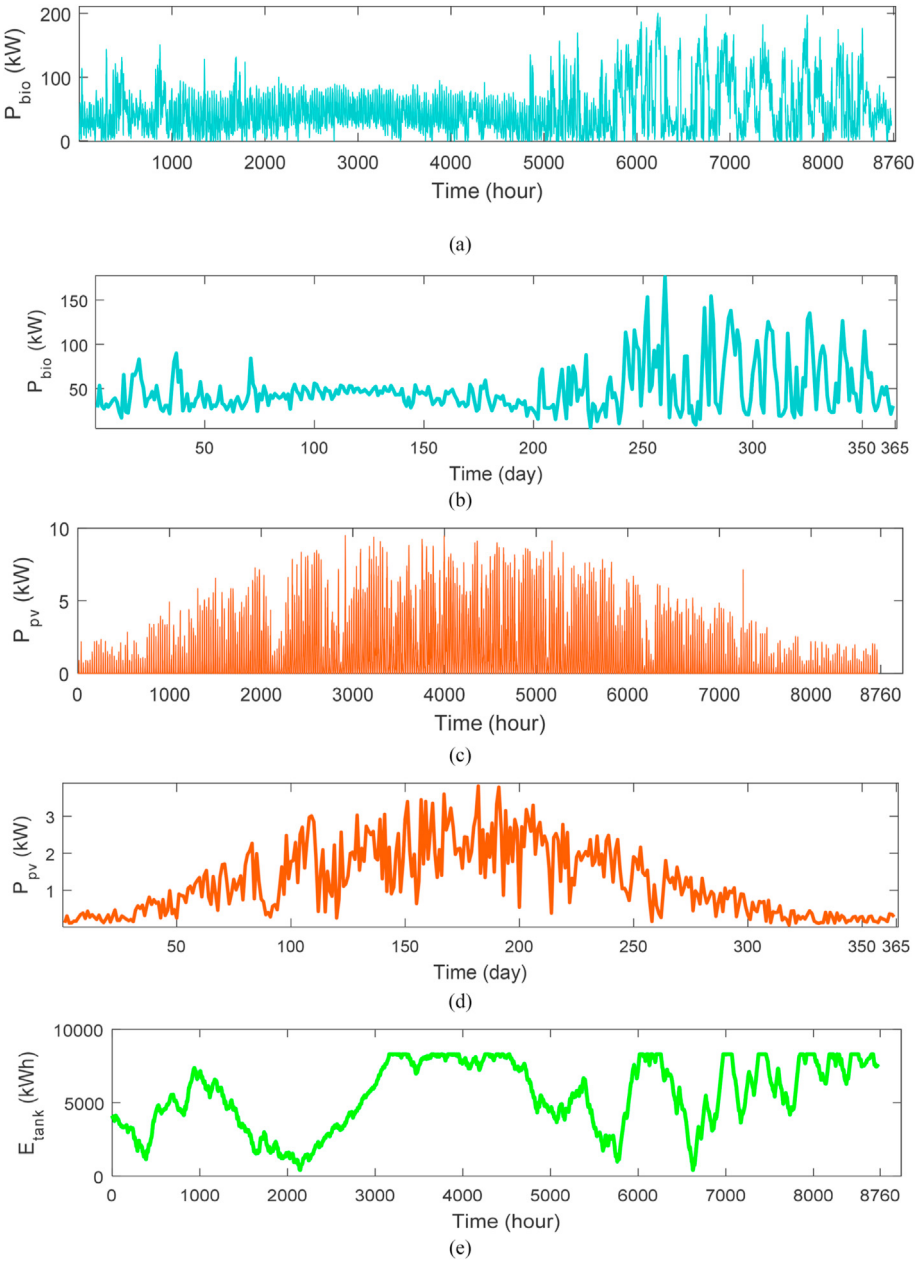
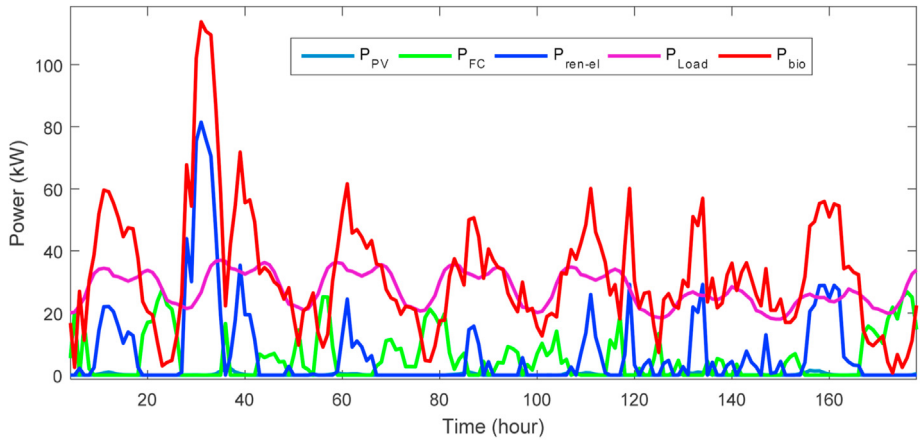


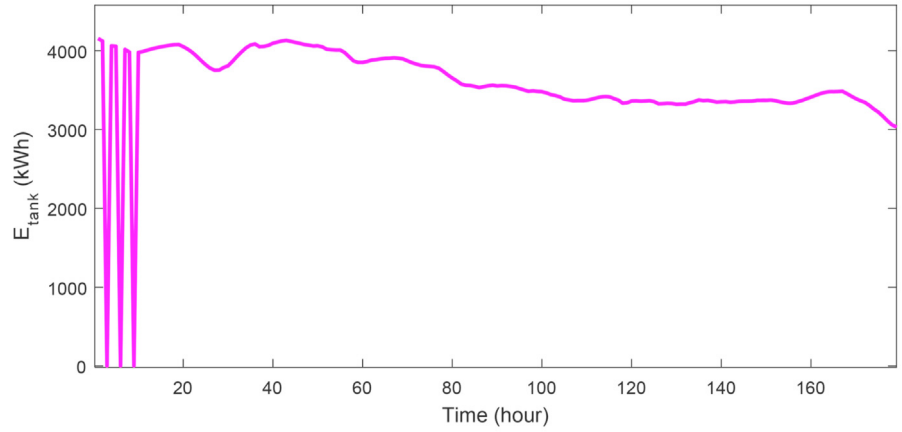
Fig. 8. Contribution of units in Scenario 2, a) Annual biowaste power b) biowaste power with an hourly average of a day c) annual PV power and d) PV power with an hourly average of a day and e) annual hydrogen energy.

equipment. Given the probability of outage of generation units, it seems logical to consider it in their probability of operation to apply a real perspective to the design of the hybrid energy systems. In the base case of the study, according to Table 2, the availabilities of all generation and storage units are assumed equal to unity without the probability of an outage. In this section, the effect of changes in the availability of PV and biowaste units on system design is investigated. To this end, the availability of PV and biowaste units is considered 0.95 % and 0.90 %, respectively. The convergence curve of the WOA algorithm in solving the optimization problem for availabilities of 0.95 and 0.9 along with the base case convergence diagram (unity availability) are depicted in Fig. 11. Also, the curve of the annual energy storage in the hydrogen tank for various availability values along with the energy curve of the tank in the base case is illustrated in Fig. 12.

Fig. 12, shows the curve of the annual energy storage in the hydrogen tank in Scenario 4. It is observed that with the reduction of the availability of generation units, the amount of energy storage in the hydrogen tank has also decreased. A comparison of cost values and reliability indices is also presented in Tables 9 and 10. The COE in PV-Biowaste-FC system designing for availability values of 1, 0.95 and 0.9 is obtained 0.5238, 0.5430 and 0.5645 $. So with decreasing the availability, the COE is increased. Also, the results showed that with decreasing the availability, the reliability indices are weakened. The reason for this is that with the reduction in the probability of operation of PV and biowaste equipment (availability reduction), these units have had to produce more power to achieve the scale of load supply in the base case, which causes the use of more units, which of course increases the TNPC of the whole system. On the other hand, it can be observed that when the



(a)



(b)

Fig. 9. The contribution of a) different units and b) hydrogen storage energy for the first week of the year in Scenario 2.

probability of availability is considered real and non-unity, in addition to increasing system costs, reliability indices are also weakened. Therefore, for energy system designers, with regard to the probability of outages of the equipment in this type of system and by considering the availability of power generating units, a more realistic view of these types of systems can be created.

6.5. Simulation results of scenario 4

In this scenario, the effect of changes in the inverter availability on system design is evaluated. In the base case scenario, the inverter availability was considered equal to unity. In this section, the effect of 99 % inverter availability on system design is investigated and the results are compared. The convergence curve of the WOA algorithm in solving the optimization problem of different inverter availability values is shown in Fig. 13. Also, the curve of annual energy storage in the hydrogen tank for different availability values is illustrated in Fig. 18. According to Fig. 13, it is observed that as the inverter availability decreases, the system cost increases significantly. An inverter is an interface device for converting DC power to AC located before the load. Any failure of the inverter will impede supplying the load. Therefore, the possibility of availability of the inverter has a great impact on the design of the hybrid system. In Fig. 14, the energy storage in the hydrogen tank is plotted for different inverter availability values. As can be seen, as the inverter availability decreases, the energy level in the tank increases.

Table 11, provides the optimal equipment sizes are given for the different inverter availability values in, and the comparison of cost values and reliability indices are given in Table 12. The results show that by reducing the inverter availability, system costs and costs incurred to supply one kWh of the demand have increased. TNPC has increased from 2.820 M$ to 3.159 M$ and COE from 0.5238 $ to 0.5868 $. Also, the LPSP of the system has increased from 0.0029 to 0.0041 and also LOEE and LOLE are weakened. Therefore, inverter availability is a very important factor in the technical and economic design of hybrid systems.

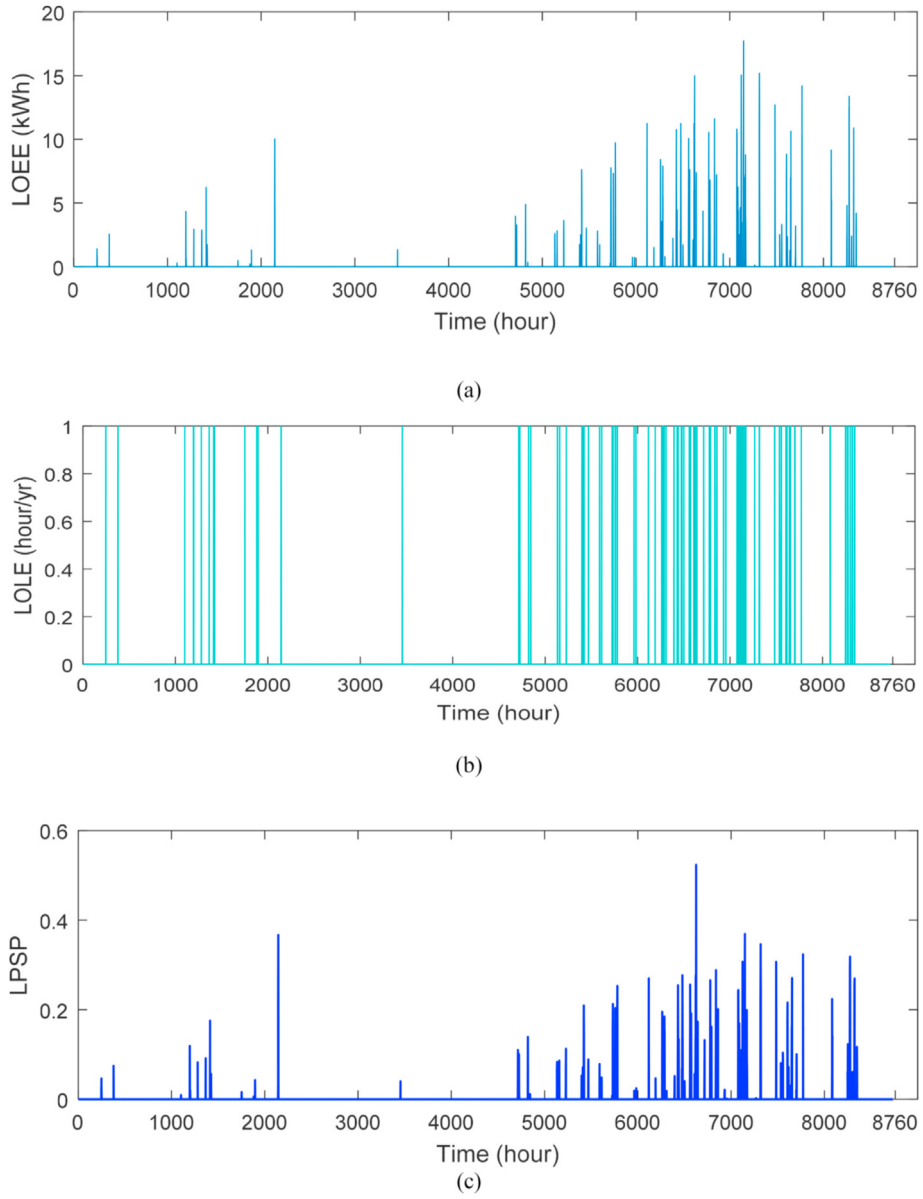


Fig. 10. The reliability indices in Scenario 2 a) LOEE b) LOLE c) LPSP.

Table 8
Comparison of cost values and reliability indices of Scenarios 1 and 2.

Scenario/Parameter	TNPC (M$)	COE ($/kWh)	LPSP	LOLE (hr/yr)	LOEE (MWh/yr)
Bio-FC) Scenario 1)	2.850	0.5294	0.0060	295	1.615
PV-Biowaste-FC) Scenario 2)	2.820	0.5238	0.0029	220	0.974

6.6. Simulation results of scenario 5

In this scenario, the effect of an increase in the investment cost of the fuel cell on the design of the hybrid system is examined. The cost has been increased from $ 3000 to $ 5000. The results obtained from the evaluation provided in Tables 13 and 14 and Fig. 15, which include the optimal size of equipment, cost values, and reliability indices.

According to the results of Tables 13 and 14, it is observed that by increasing the initial investment cost of the fuel cell, the

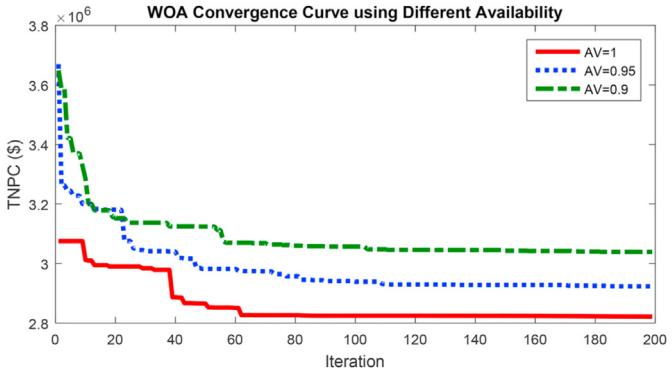


Fig. 11. Comparing the convergence curve of the WOA method for different availability values.

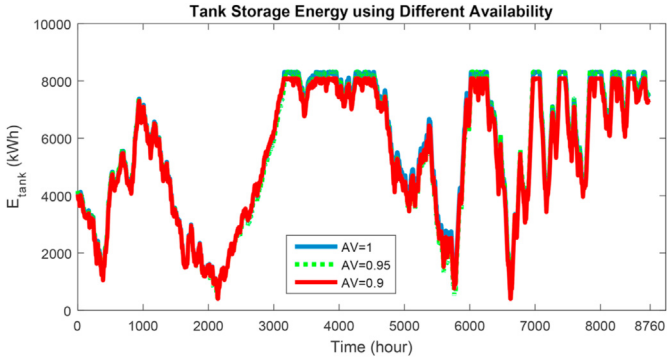


Fig. 12. The curve of the annual energy storage in the hydrogen tank for different availability values.

Table 9
Comparing the optimal size of the system equipment for different availability values.

Parameter	$N_{Biowaste}$	A_{PV}	N_{el}	$N_{\tan k}$	N_{FC}	N_{inv}
AV = 1	186	78.62	61	209	31	47 (47.47)
AV = 0.95	196	62.44	63	207	31	47 (47.18)
AV = 0.9	207	86.22	60	203	32	47 (47.09)

Table 10
Comparing costs and reliability indices for different availability values.

Parameter	TNPC (M$)	COE ($/kWh)	LPSP	LOLE (hr/yr)	LOEE (MWh/yr)
AV = 1	2.820	0.5238	0.0029	220	0.974
AV = 0.95	2.923	0.5430	0.0032	241	1.060
AV = 0.9	3.039	0.5645	0.0036	250	1.176

program decreases power generation by the fuel cell, and to prevent the increase in the system costs, increases power generation by PV panels. The results also indicate that the increase in the initial investment cost of the fuel cell, which has led to a reduction in power generation, has weakened the reliability indices of the load. The reason is that the storage system plays a very important role in providing continuous and sufficient supply. Figs. 15–18, shows the generated power of PV panels, biowaste, fuel cell and loss of power

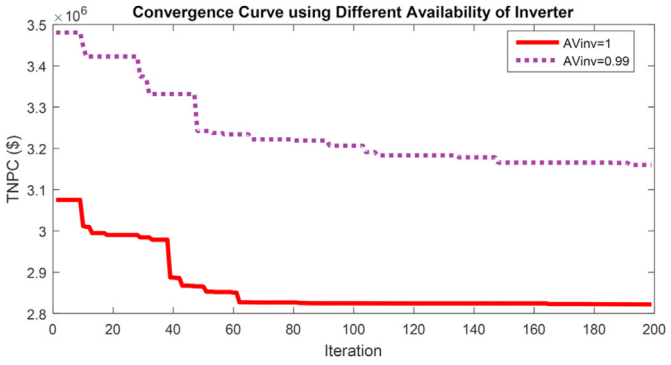


Fig. 13. Comparing the convergence curve of the WOA method for different availability values.

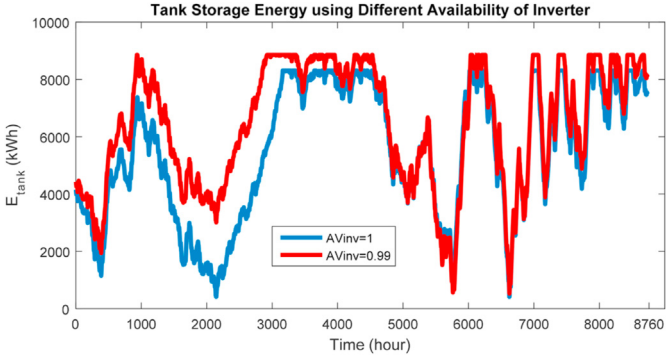


Fig. 14. The curve of the annual energy storage in the hydrogen tank for different availability values.

Table 11
Comparing the optimal size of the system equipment for different inverter availability values.

Parameter	$N_{Biowaste}$	A_{PV}	N_{el}	$N_{\tan k}$	N_{FC}	N_{inv}
AV_{inv} = 1	186	78.62	61	209	31	47 (47.47)
AV_{inv} = 0.99	193	14.71	60	223	51	47 (47.29)

supply probability for initial investment cost on fuel cell equal to $ 3000 and $ 5000 during a year. As can be seen, with the increase in the initial investment cost of the fuel cell as a storage system, the power generation by the fuel cell has decreased, while it has increased for PV panels. It is also observed that with the increase in the initial investment cost of the fuel cell, in addition to increasing the total energy generation costs of the system (refer to Table 14), the LPSP index value has been weakened. Thus, the increase in $C_{cap,FC}$ from $ 3000 to $ 5000 reduces fuel cell participation and weakens reliability.

6.7. Results comparison

In this section, the results of all scenarios are evaluated together in terms of the TNPC and LPSP according to Figs. 19 and 20. As can be seen, the TNPC and LPSP of the second scenario is lower than

Table 12
Comparing costs and reliability indices for different inverter availability values.

Parameter	TNPC (M$)	COE ($/kWh)	LPSP	LOLE (hr/yr)	LOEE (MWh/yr)
$AV_{inv} = 1$	2.820	0.5238	0.0029	220	0.974
$AV_{inv} = 0.99$	3.159	0.5868	0.0041	232	2.68

Table 13
Comparing the optimal size of the system equipment for different inverter availability values.

Parameter	$N_{Biowaste}$	A_{PV}	N_{el}	$N_{\tan k}$	N_{FC}
$C_{cap,FC} = 3000$	186	78.62	61	209	31
$C_{cap,FC} = 5000$	184	113.93	59	199	29

other scenarios. In other words, the contribution of PV and Biowaste resources together with the FC storage system has resulted in a lower TNPC and LPSP. On the other hand, it is observed that the decrease in the availability of generation resources and inverters (in scenarios 3 and 4) as well as the increase in the investment cost of the storage system, increase the TNPC and LPSP in comparison with the second scenario.

Moreover, the COE is considered as a measure of the potential and feasibility of implementing hybrid energy systems in different areas. Therefore, less or more of this cost does not in any way imply the superiority or weakness of the proposed method in a different area, but rather the potential evaluation of that region in view of the proposed hybrid system generation cost. In this section, the COE of hybrid PV-Biowaste-FC system designing based WOA is compared with previous studies that are presented in Table 15. The COE refers to the cost of energy per kW that is taken for previous hybrid system designing researches in different regions. The COE value can vary according to the geographical location of the region and the potential of the input data such as irradiance and wind speed. A low value of COE means less cost per kWh of load supply and vice versa. As can be seen, the COE value for the proposed hybrid system (PV-Biowaste-FC) is $ 0.5238. It is cleared that for different regions presented in Table 14, lower COE by PV-Biowaste-FC, and indicates that the radiation potential is very good in this region. Compared to these regions studied in previous researches, the cost per kWh of load supply based on PV-Bio hybrid system is less than the other regions with different hybrid systems.

7. Conclusion

A designing framework of a hybrid PV-Biowaste-FC system with hydrogen storage was presented considering the availability of components using the WOA algorithm. The objective function was considered as minimizing the TNPC during the 20-year useful lifetime of the system with satisfying the LPSP and LOLE. The optimal size of the system components is determined optimally using the WOA. The results showed that the hybrid PV-Biowaste-FC system was achieved as an optimal combination with minimum TNPC and better reliability indices. The results also demonstrated that the contribution of the photovoltaic and biowaste resources together with hydrogen storage can be reduced more the system TNPC in comparison with the hybrid biowaste-FC system. In addition, the results also confirmed the superiority of the WOA

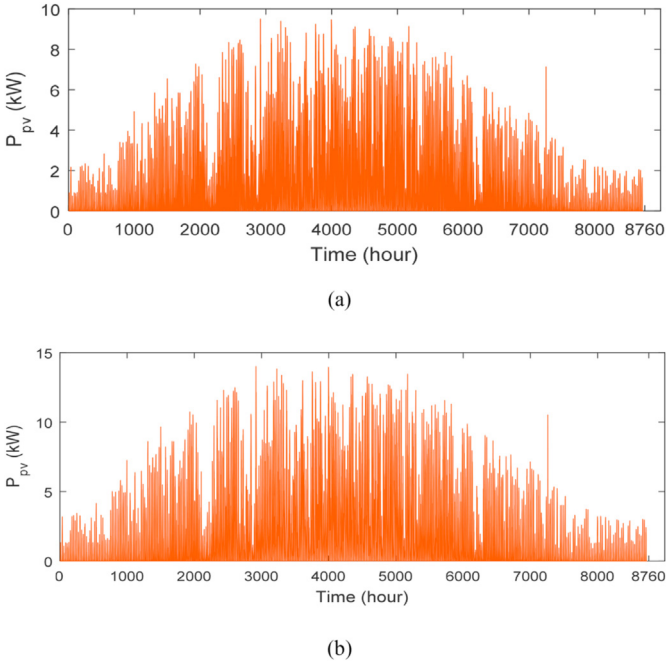


Fig. 15. The PV generated power for a) $C_{capital,FC} = 3000\$$ and b) $C_{capital,FC} = 5000\$$.

Table 14
Comparing costs and reliability indices for different inverter availability values.

Parameter	TNPC (M$)	COE ($/kWh)	LPSP	LOLE (hr/yr)	LOEE (MWh/yr)
$C_{cap,FC} = 3000$	2.820	0.5238	0.0029	220	0.974
$C_{cap,FC} = 5000$	2.890	0.5368	0.0052	319	1.68

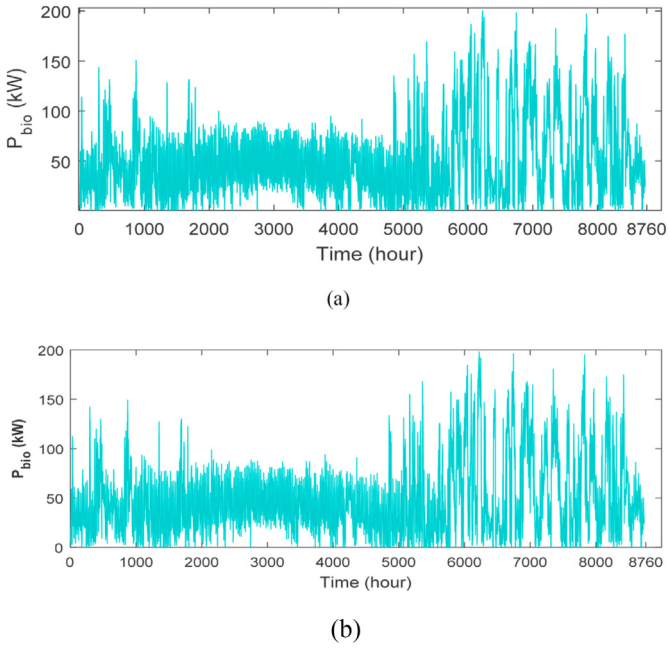


Fig. 16. The biowaste generated power for a) $C_{capital,FC} = 3000\$$ and b) $C_{capital,FC} = 5000\$$.

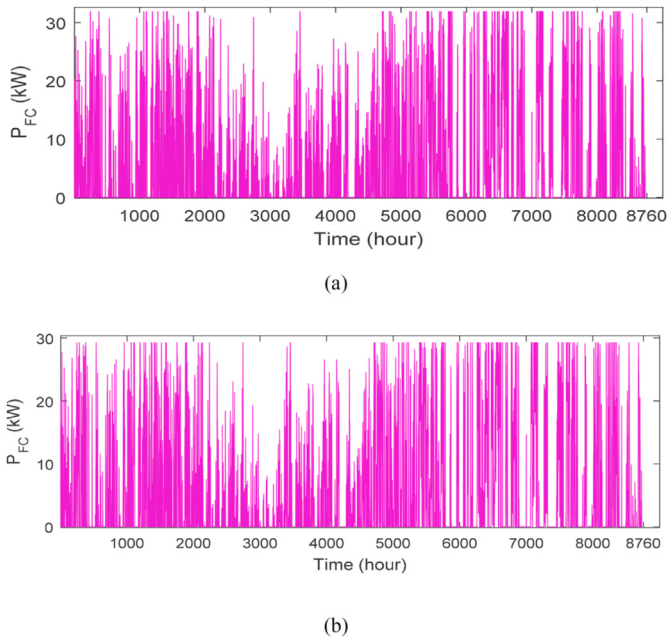


Fig. 17. The FC generated power for a) $C_{capital,FC} = 3000\$$ and b) $C_{capital,FC} = 5000\$$.

compared with the PSO in obtaining lower TNPC and better reliability indices. The results also cleared that the energy storage level and the TNPC was decreased with increasing the generation resources availability and the reliability indices were also improved in this condition. Also, the results showed that the TNPC was decreased and the reliability indices were improved with increasing the inverter availability. Moreover, the effect of changes in fuel cell investment cost was investigated on the hybrid system designing. The results demonstrated that in the condition of

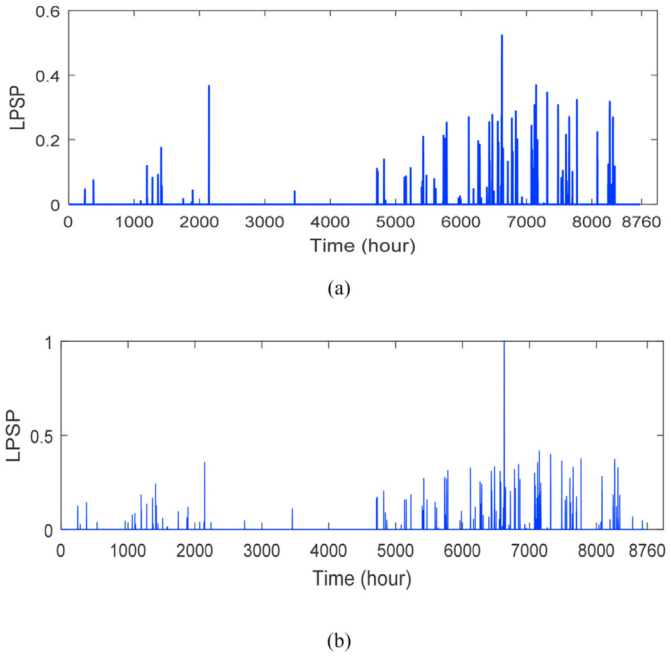


Fig. 18. The LPSP generated power for a) $C_{capital,FC} = 3000\$$ and b) $C_{capital,FC} = 5000\$$.

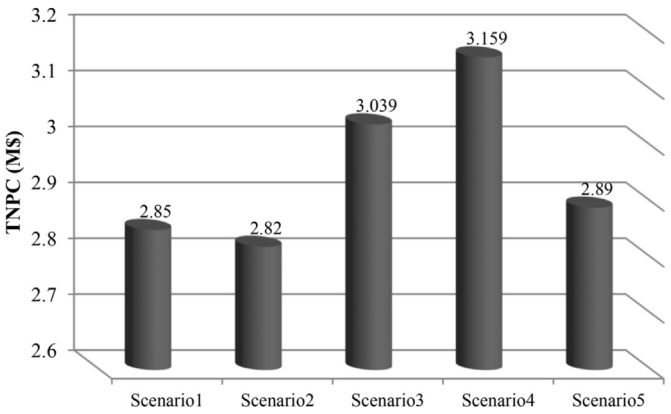


Fig. 19. The results comparison of the different scenarios in view of the TNPC.

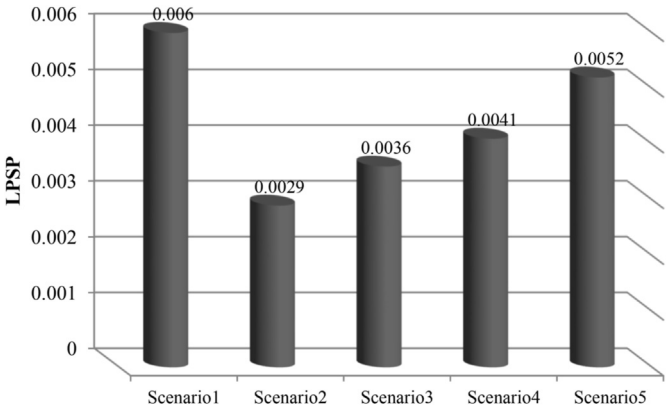


Fig. 20. The results comparison of the different scenarios in view of the LPSP.

Table 15
Comparison with previous studies.

Hybrid system	Region	COE ($/kWh)
PV-Biowaste-FC	Ardabil, Iran	0.5238
PV-WT-Diesel-Battery [29]	Nahavand, Iran	1.87
PV-Wind-FC [30]	Istanbul, Turkey	0.837
PV-WT-FC [18]	Gonbad, Iran	0.87
PV-Wind-FC [31]	Tehran, Iran	0.83
PV-FC-Battery [32]	Jeddah, Saudi Arabia	1.40
PV-Wind-FC [19]	Gorgan, Iran	0.853

increasing the fuel cell investment cost, fuel cell power was reduced, photovoltaic power was increased, the system TNPC was increased and also the reliability indices were weakened. Access to hourly data of irradiance and temperature were the main limitations of the study. The hybrid system designing as multi-criteria optimization considering demand uncertainty is suggested for future work.

Authorship contributions

Hongyue Sun: Methodology. Abdol Ghaffar Ebadi: Resources. Mohsen Toughani: Visualization. Saber Arabi Nowdeh: Data curation. Amirreza Naderipour: Writing – original draft. Aldrin Abdullah: Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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